

Big Data Storage

BDS – Curricular Unit File 2025-2026

Lecturer

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Office hours

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General objectives

Today, data is everywhere and comes in different shapes and volumes, in structured and unstructured forms. Big Data is a field dedicated to the analysis, processing, and storage of large collections of data that frequently originate from disparate sources. Big Data solutions are required when traditional data storage technologies are insufficient. This course covers the essentials of Big Data from a **technical and business** perspective. The objective of the course is to familiarize students with the key concepts and terminology related to Big Data storage (Sharding, clusters, NoSQL, CAP theorem, among others), learn from business cases that applied Big Data, and learn the fundamentals of MongoDB, a document database technology. The students will have labs on how to set up a MongoDB database and explore different ways to search, create, and analyze data. Lastly, we will cover database performance basics and discover how to get started with developing applications that use MongoDB.

Sustainable development goals (SDG)

8 - Promover o crescimento económico inclusivo e sustentável, o emprego pleno e produtivo e o trabalho digno para todos

11 - Tornar as cidades e as comunidades mais inclusivas, seguras, resilientes e sustentáveis

16 - Promover sociedades pacíficas e inclusivas para o desenvolvimento sustentável, proporcionar o acesso à justiça para todos e construir instituições eficazes, responsáveis e inclusivas a todos os níveis

Enrollment Requirements

The lectures, group discussions, reading materials, quizzes, and exams will be all in English language. Although not mandatory, the students should be proficient in English language.

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Learning outcomes of the curricular unit (LO)

Upon completion of this course, students should be able to:

- LO1.** Describe Big Data characteristics and key terminology.
- LO2.** Differentiate between structured and unstructured data.
- LO3.** Explain business drivers for Big Data adoption.
- LO4.** Identify key planning considerations in Big Data initiatives.
- LO5.** Outline enterprise technologies and BI tools used in Big Data.
- LO6.** Understand sharding, replication, CAP theorem, ACID, and BASE.
- LO7.** Compare on-disk and in-memory storage for data processing.
- LO8.** Recognize NoSQL database types and appropriate use cases.
- LO9.** Discuss the role of Big Data in driving business value.
- LO10.** Understand MongoDB fundamentals and Atlas features.
- LO11.** Import, export, and query data in MongoDB.
- LO12.** Perform advanced CRUD operations in MongoDB.
- LO13.** Apply indexing and aggregation pipelines in MongoDB.

Syllabus

Lessons

- LU1.** Introduction to the course, key terminology, and characteristics of Big Data.
- LU2.** Business motivation, architecture, and drivers for Big Data adoption. Big Data adoption planning and analytics lifecycle.
- LU3.** Big Data analytics lifecycle (continuation). Data warehousing and data lakes.
- LU4.** Enterprise technologies and Big Data Business Intelligence (OLTP, ETL, Data Marts). Big Data storage concepts: sharding and replication.
- LU5.** Big Data storage concepts: ACID, CAP, BASE. Introduction to NoSQL databases.
- LU6.** Introduction to MongoDB. Installation using Atlas UI and Compass. Converting a CSV file to a JSON file.
- LU7.** Document databases and vector search in MongoDB.
- LU8.** Importing, exporting, and querying data.
- LU9.** Creating and manipulating documents.
- LU10.** Advanced CRUD operations.
- LU11.** Vector search exercise. Indexing and aggregation pipeline.
- LU12.** Project support and defense – Session I.
- LU13.** Project support and final defense – Session II.

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Curricular unit Planning

Week	Theory	Labs
1	1. Rules of course and Key terminology. 2. Big data characteristics	Review of Big Data tools (installation and test)
2	3. Business Motivation, architecture, and drivers of Big data adoption. 4. Big Data Adoption, Planning Considerations, and big data analytics lifecycle	Cloud Platforms for Big Data, basic setup and processing
3	5. Big Data Analytics Lifecycle. Continuation	Data Warehousing and Data Lakes
4	6. Enterprise Technologies and Big Data Business Intelligence (OLTP, ETL, Data Marts) 7. Big Data Storage Concepts (Sharding, Replication).	Data Ingestion and Integration (ETL)
5	8. Big Data Storage Concepts (ACID, CAP, BASE) 9. NoSQL databases.	Introduction to MongoDB. MongoDB Installation: the UI Atlas. MongoDB Compass and Convert a CSV file to a JSON file
6	10. Document databases (MongoDB) 11. Vector search (MongoDB).	Importing, Exporting, and Querying Data - Part I. Importing, Exporting, and Querying Data - Part II
7	12. Presentations 13. Presentations	Creating and Manipulating Documents - Part I. Creating and Manipulating Documents - Part II
8	14. Presentations	Advanced CRUD Operations - Part I. Advanced CRUD Operations - Part II
9	Not applicable	Vector search exercise
10	Not applicable	Indexing and Aggregation Pipeline - Part I
11	Not applicable	Indexing and Aggregation Pipeline - Part II
12	Not applicable	Project support and defense
13	Not applicable	Project support and defense
14	Not applicable	Project defense

Demonstration of the syllabus coherence with the curricular unit's learning objectives

The learning units (LU) cover the learning outcomes (LO) as follows:

- LO1 is addressed in the LU1.
- LO2 is addressed in the LU2.
- LO3 is addressed in the LU3.
- LO4 is addressed in the LU4.
- LO5 is addressed in the LU5.
- LO6 is addressed in the LU6.
- LO7 is addressed in the LU7.
- LO8 is addressed in the LU8.
- LO9 is addressed in the LU9.
- LO10 is addressed in the LU10.
- LO11 is addressed in the LU11
- LO12 is addressed in the LU12.
- LO13 is addressed in LU13.

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Teaching methodologies

The course combines theoretical, labs, and case study presentations and discussions. The theoretical component includes lectures and slide show demonstrations to present concepts and discuss problems in class. The labs include individual hands-on exercises that are applicable to the final project, where the students will be able to apply all the knowledge acquired in class. Moreover, the lessons apply short quizzes throughout the semester to assess the student's continuous progress and understanding of the contents.

Evaluation

Regular examination period (1st epoch)

- Final exam (40%).
- Group presentation in theoretical class (10%)
- Practical class component (50%) (the teacher of practical class is free to distribute this 50% in one or various components).
- Bonus! Group participation during theoretical class (5%).

Resit examination period (2nd epoch)

- Same as for the first epoch

Special epoch: Exam 100%

Rules:

- Minimum score in the exam: 9.5 out of 20.
- The teams are made up of 4 to 5 students.
- Project delays are penalized with 1 point per day up to a maximum of 5 points.

Demonstration of the coherence between the teaching methodologies and the learning outcomes.

The theoretical lessons provide the students with the theory about core concepts about the big data storage technologies and MongoDB document database. The labs will allow the students to put into practice what they have learned in the theoretical classes with the support in-situ of the lab teachers. And finally, the students will cement the knowledge from theoretical and practical lessons by implementing a project and answering short quizzes.

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Recommended Bibliography

- Erl, T., Khattak, W., & Buhler, P. (2016). Big data fundamentals: concepts, drivers & techniques. Prentice Hall Press.
- Fei Hu (2016). Big Data: Storage, Sharing, and Security 1st Edition
- Joe Reis, Matt Housley (2022). Fundamentals of Data Engineering: Plan and Build Robust Data Systems 1st Edition
- Gerardus Blokdyk (2021). Big Data Storage A Clear and Concise Reference
- Gerardus Blokdyk (2018). Big Data Storage Solutions Third Edition
- Lemahieu, W., Vanden Broucke, S., & Baesens, B. *Principles of Database Management: The Practical Guide to Storing, Managing and Analyzing Big and Small Data*. Cambridge University Press (2018).
- Bradshaw, S., Brazil, E., & Chodorow, K. (2019). *MongoDB: the definitive guide: powerful and scalable data storage*. O'Reilly Media.
- Phaltankar, A., Ahsan, J., Harrison, M., & Nedov, L. (2020). MongoDB fundamentals: a hands-on guide to using MongoDB and atlas in the real world.
- Marr, B. (2016). *Big data in practice: how 45 successful companies used big data analytics to deliver extraordinary results*. John Wiley & Sons.

Software

2. MongoDB Atlas and Compass